

$$= (9/\sqrt{2})/20 = 2.0 \text{ W}$$

27 D Total force exerted due to reflection of photons $F = \frac{N(2mc)}{t}$

But intensity of light $I = \frac{Nhf}{tA}$, so $\frac{N}{t} = \frac{IA}{hf}$

Also $E = mc^2 = hf$

Total force $F = \frac{IA}{hf}(2mc) = \frac{IA}{mc^2}(2mc) = \frac{2IA}{c}$

28 C Since $\Delta E = hf$, from energy level diagrams, the energy difference is lesser towards